# Lucas' Theorem &

PS Study Notes

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# 1

$N$ , $N+1$

$$\binom{N}{0},\ \binom{N}{1},\ \cdots,\ \binom{N}{N}$$

.

- $0 \le N \le 10^{18}$
- 100,000

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# 2 : Lucas' Theorem

## 2.1 Statement

$p$ , $N$ $k$ $p$ :

$$\binom{N}{k} \equiv \prod_i \binom{n_i}{k_i} \pmod{p}$$

$N = \sum n_i p^i$, $k = \sum k_i p^i$.

## 2.2 p = 2 ( )

$\binom{0}{1} = 0$ :

$\binom{N}{k}$ $\iff$ $k$ $N$ $\iff$ $k$ $N$ **binary submask**

### 2.3 Submask ?

`(k & N) == k`  $k$ $N$ submask .
, $k$  1  $N$  1 .

```
N = 1010   k = 0010  →  (k & N) = 0010 = k  →  submask    →
N = 1010   k = 0001  →  (k & N) = 0000   k  →  not submask  →
```

### 2.4 (GF(2))

GF(2)  Freshman's dream:

$$(1+x)^{2^i} \equiv 1 + x^{2^i} \pmod 2$$

$N = \sum_i n_i \cdot 2^i$  :

$$(1+x)^N = \prod_{i:\ n_i=1} (1+x)^{2^i} \equiv \prod_{i:\ n_i=1} (1+x^{2^i}) \pmod 2$$

$x^k$  $x^{2^i}$  → submask .

---

## 3

$$= 2^{\mathrm{popcount}(N)}$$

$$\boxed{= (N+1) - 2^{\mathrm{popcount}(N)}}$$

popcount$(N)$ $N$  1 .

| $N$ | $N$ ( ) | popcount | $(2^p)$ | $(N+1)$ | |
|---|---|---|---|---|---|

### 3.1

| $N$ | $N$ ( ) | popcount | $(2^p)$ | $(N+1)$ | |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 1 | 0 |
| 1 | 1 | 1 | 2 | 2 | 0 |
| 4 | 100 | 1 | 2 | 5 | 3 |
| 10 | 1010 | 2 | 4 | 11 | 7 |
| 100 | 1100100 | 3 | 8 | 101 | 93 |

---

## 4

### 4.1

```
#include <stdio.h>
using ull = unsigned long long int;

//     popcount (Hamming Weight    )
ull popcount_sw(ull x) {
    x = x - ((x >> 1) & 0x5555555555555555ULL); // 2
    x = (x & 0x3333333333333333ULL)
      + ((x >> 2) & 0x3333333333333333ULL);      // 4
    x = (x + (x >> 4)) & 0x0F0F0F0F0F0F0F0FULL; // 8
    return (x * 0x0101010101010101ULL) >> 56;    //
}

int main(void)
{
    int t; scanf("%d", &t);
    for (int tc = 1; tc <= t; ++tc)
    {
        ull n; scanf("%llu", &n);

        ull p       = popcount_sw(n);  // N
```

```
        ull odd_cnt = 1ULL << p;        //        = 2^p
        ull ans     = (n + 1ULL) - odd_cnt;

        printf("#%d %llu\n", tc, ans);
    }
}
```

### 4.2

| | | | |
|---|---|---|---|
| (k=0~N ) | | $O(N)$ per query | $N \le 10^{18} \rightarrow$ |
| Lucas | + popcount | $O(1)$ per query | POPCNT = 1 |

---

## 5 Popcount

### 5.1 (x86-64 POPCNT)

`__builtin_popcountll` , POPCNT CPU .

```
POPCNT rax, rdi    ; 1  ,  O(1)
```

### 5.2 (Hamming Weight)

POPCNT bit-parallel . $\log_2 W = 6$ (64 )

```
 :    1 0 1 1 0 1 0 0   ( : 8 )
        ↓
 1:  01 10 01 00        (2     )
        ↓
 2:  0011 0001          (4     )
        ↓
 3:  00000100           ( : 1    = 4)
```

. 64 .

### 5.3 O(1)

| | | |
|---|---|---|
| POPCNT | $O(1)$ | 1 |
| Hamming Weight | $O(\log W) = O(1)$ | ~6 |
| | $O(W)$ | 64 |

$W = 64$ $O(\log W)$ $O(1)$ .

# 6

## 6.1 1: popcount

```
//  : popcount
ull ans = popcount_sw(n);
printf("#%d %llu\n", tc, n + 1 - ans);   // ←

//  : 2^popcount
ull odd_cnt = 1ULL << popcount_sw(n);
ull ans     = (n + 1ULL) - odd_cnt;       // ←
```

## 6.2 2:

```
//  : 1 << p   p >= 32  int
ull odd_cnt = 1 << p;

//  :   1ULL
ull odd_cnt = 1ULL << p;
```

## 6.3 3:

```
//  : ans  ull  %d
printf("#%d %d\n", tc, ans);

//
printf("#%d %llu\n", tc, ans);
```

### 6.4 4: n <= 1

```
//        (      )
if (n <= 1) { printf("#%d 0\n", tc); continue; }
// n=0: 2^0=1, ans=1-1=0
// n=1: 2^1=2, ans=2-2=0
```

---

# 7

## 7.1 Kummer's Theorem

$\binom{m+n}{m}$ $p$ $= p$ $m+n$ (carry) .

Lucas Kummer ( ) .

## 7.2 p

$p$ , $k_i > n_i$ :

$$\binom{N}{k} \equiv 0 \pmod{p}$$

$O(\log_p N)$ per query.

## 7.3

- **BOJ 11401** — mod $p$ ( + )
- **BOJ 16977** —